A COMPARISON OF LEXICAL AND CONTEXTUAL EMBEDDING REPRESENTATIONS FOR JOB POSTING ANALYSIS

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TEXT MINING AND SEARCH
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THE FRAMEWORK

The rapid growth of online job postings makes labor market analysis a relevant and data-rich application of text mining techniques.

THE PROBLEM

Job descriptions contain valuable semantic information, but their unstructured and heterogeneous nature makes automatic analysis challenging.

RESEARCH QUESTIONS

- Can job descriptions be automatically classified to accurately predict the corresponding job position?
- Is it possible to identify latent topics that capture shared semantic patterns across different job descriptions using unsupervised methods?

STRUCTURE OF THE DATASET

The dataset is composed of **30,570** online job advertisements, representing a broad and heterogeneous snapshot of the labor market.

Each job posting contains several attributes, but for the purpose of this study only the **job title** and the **job description** were retained, as they concentrate most of the relevant semantic information.

After **removing duplicate** entries, the final dataset includes **615 distinct job positions**, covering a wide range of professional profiles.

The resulting class distribution is slightly **unbalanced**, a characteristic that mirrors realistic hiring scenarios and increases the complexity of the classification task.

PREPROCESSING STRATEGY

A task-aware preprocessing strategy was adopted to avoid artificial bias and to preserve the most relevant linguistic information.
Preprocessing choices were therefore adapted to the specific requirements of the downstream models.

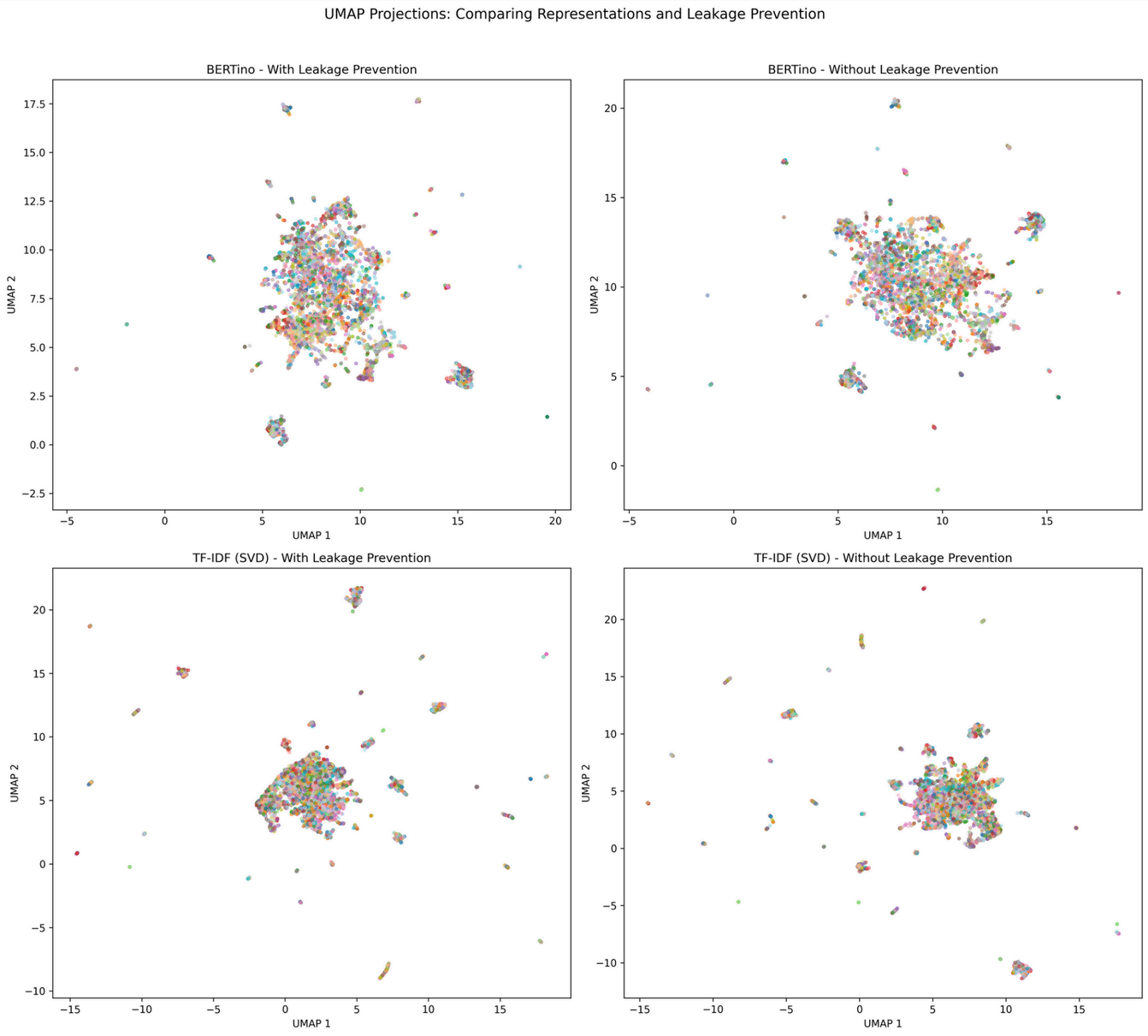
Aggressive cleaning for Bag-of-Words models

- Removal of job titles
- Stopwords
- Non-alphabetic character filtering
- Lowercasing and vocabulary cleaning

Minimal preprocessing for contextual embeddings

- Removal of job titles

SCATTER PLOT UMAP TF-IDF VS BERTINO WITH AND WITHOUT LABEL LEAKAGE



TEXT REPRESENTATION

Two complementary text representations were adopted to capture both lexical and semantic information in job descriptions.

TF-IDF

- Sparse, high-dimensional representation based on term frequency
- Similarity driven by shared vocabulary and domain-specific keywords
- Highly interpretable and effective with lexical models

BERTino

- Dense contextual embeddings capturing semantic information
- 768-dimensional vectors
- Groups documents with similar meaning despite different vocabularies
- Less interpretable and characterized by complex, non-linear geometry

TEXT CLASSIFICATION

The classification task involves a challenging **multi-class setting** with many semantically overlapping job positions. Results show that **TF-IDF** combined with non-linear classifiers outperforms contextual embeddings, highlighting the importance of domain-specific keywords. It is also possible to notice that:

- Best performance achieved by TF-IDF with Random Forest
- Embedding-based models struggle with class separability
- Errors mainly caused by vague or overlapping job descriptions
- Confusions often occur between closely related roles or contract types

Model	Accuracy
TF-IDF + LogReg	0.5226
TF-IDF + RF	0.6618
TF-IDF + SGD	0.4850
BERTino + RF	0.5864
BERTino + LogReg	0.3448
BERTino + SGD	0.2562

TOPIC MODELING

Topic modeling was used as an unsupervised approach to explore latent semantic structures in job descriptions to find recurrent themes in insertions.
Different statistical and embedding-based models were compared, using a fixed number of **18 topics** estimated through HDP, in order to analyze the trade-off between topic interpretability and separation. The models chosen for this tasks are:

1. LSA
2. LDA
3. HDP
4. BERTopic

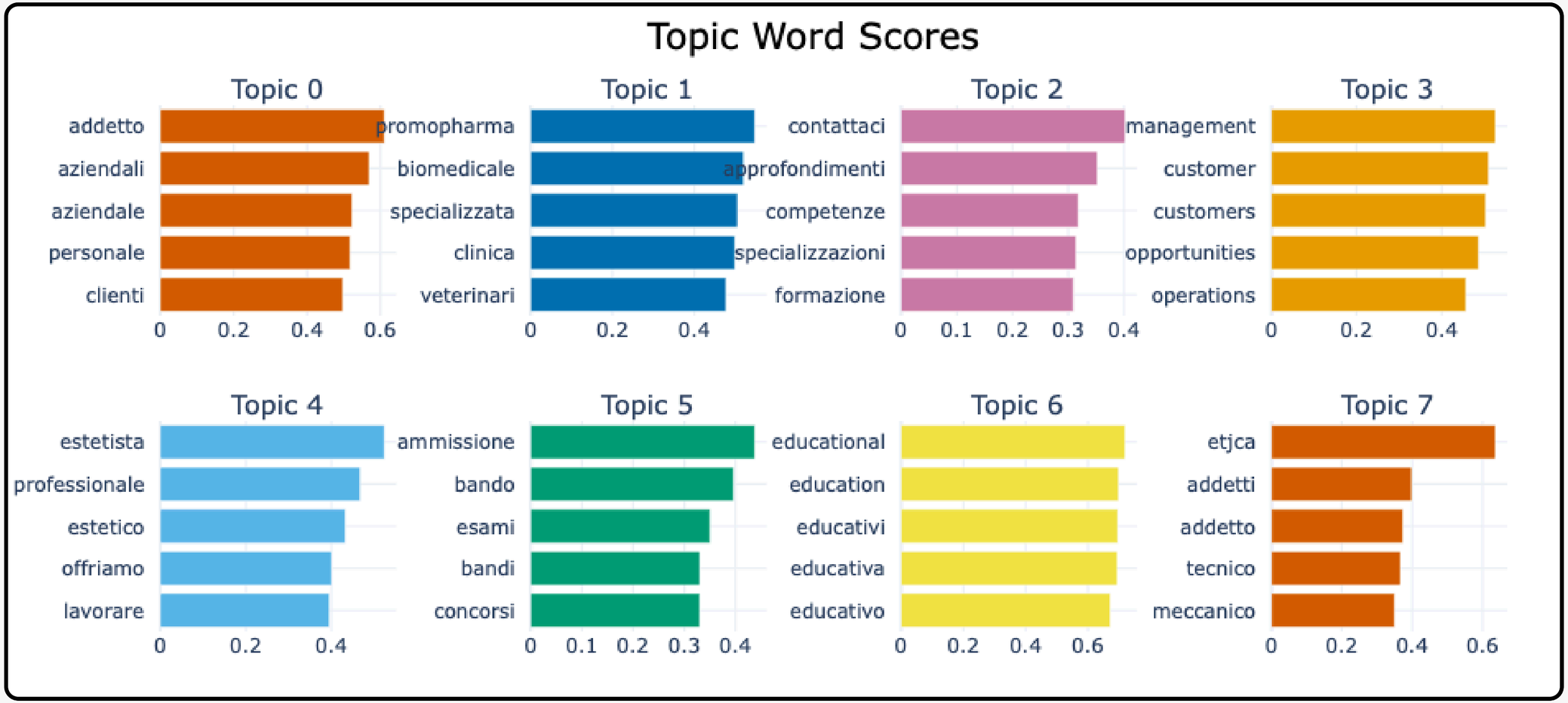
TOPIC MODELING

Topic quality was evaluated using **coherence**, **diversity**, and **perplexity** when applicable. The results highlight a clear **trade-off** between internal topic coherence and separation across topics.

- **HDP** achieves the highest **coherence** but shows topic redundancy
- **BERTopic** maximizes **diversity** but produces less coherent topics
- **LDA** provides the **best balance** between interpretability and separation

Model	Coherence	Perplexity	Diversity
LSA	0.5398	/	0.5550
LDA	0.6236	-8.5202	0.7900
HDP	0.6493	-14.0123	0.6350
BERTopic	0.5509	/	0.8421

TOPIC MODELING - RESULTS



CONCLUSION

Results show that textual representation plays a crucial role in model performance and interpretability.

- For classification, **TF-IDF combined with non-linear models** outperformed contextual embeddings, indicating that domain-specific keywords are highly informative in this setting.
- In topic modeling, probabilistic approaches such as **LDA** achieved the **best balance** between coherence and diversity.

Overall, the study highlights that **simpler methods** can remain **effective** for real-world, heterogeneous textual data.

THANK YOU !